



Department of Mathematics
IIT Hyderabad
MA1110: Calculus I

August 29, 2026

Credits: 1
Fall 2026

Time: 120 Minutes

Final Exam

Maximum Marks: X

Name:

Roll No:

Instructions:

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1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive continuous function, that is, $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Then
 - (A) $f(x) = x^2$ for all $x \in \mathbb{R}$.
 - (B) There exists $\lambda \in \mathbb{R}$ such that $f(x) = \lambda x$ for all $x \in \mathbb{R}$.
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TRUE:					FALSE:				
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+4/

TRUE:	☐	☒	☐	☒	FALSE:	☒	☐	☒	☐
		B		D		A		C	
		✓		✓		✓		✓	



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+4/.

TRUE:	B	D	—	—	FALSE:	A	C	—	—
	✓	✓				✓	✓		



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+3/.

TRUE:	☒	☒	☒	☒	D	FALSE:	A	C	☒	☒
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TRUE:	B	—	D	—	FALSE:	B	—	A	C
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For both places
-1

+2/.



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TRUE:	C	—	—	—	FALSE:	C	—	—	—
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-1/2

For both the places.
-1



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+2/.

TRUE:	☒	☒	☒	☒	FALSE:	☒	☒	☒	☒
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✓

✓



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TRUE:	B	—	—	—	FALSE:	D	A	C	—
	✓					×	✓	✓	

+2½/.



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-2/

TRUE:	A	—	C	—	FALSE:	—	B	—	D
	x		x			x		x	



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0/

TRUE:	☒	☒	☒	☒	FALSE:	☒	☒	☒	☒
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TRUE:							FALSE:		
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$$f(n) = n f(1) \quad \forall n \in \mathbb{N}.$$

$$\left[\begin{aligned} f(0) &= f(0) + f(0) \\ \Rightarrow f(0) &= 0 = 0 \cdot f(1). \end{aligned} \right.$$

$$\left[\begin{aligned} f(x-x) &= f(x) + f(-x) \\ \Rightarrow f(x) &= -f(-x) \end{aligned} \right.]$$

Now, check $f(q) = q f(1) \quad \forall q \in \mathbb{Q}.$

$$1 = \underbrace{\frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}}_{n \text{ times}}.$$

Take $r \in \mathbb{Q}^c$, $x_n \in \mathbb{Q}$ and $x_n \rightarrow r.$

$$\text{So, } f(x_n) = x_n f(1) \rightarrow r f(1)$$

$$\downarrow$$

$$f(r).$$

$$\Rightarrow f(x) = x f(1) \quad \forall x \in \mathbb{R}.$$

True

B, D

False

A, C.

2. Let $f : [0, 2] \rightarrow \mathbb{R}$ be continuous on $[0, 2]$ and differentiable on $(0, 2)$, satisfying $f(0) = 0$, $f(1) = 4$, $f(2) = 2$. Then
- (A) There exists $c_1 \in (0, 1)$ such that $f'(c_1) = 4$.
 - (B) There exists $c_2 \in (1, 2)$ such that $f'(c_2) = -2$.
 - (C) There exists $c \in (0, 2)$ such that $f'(c) = 0$.
 - (D) There exist distinct $c_1, c_2 \in (0, 2)$ such that $f'(c_1) = f'(c_2) = 0$.

True A, B, C

False D.

$$\begin{array}{|l} \text{MVT} \\ \frac{f(1) - f(0)}{1 - 0} = f'(c_1) \\ c_1 \in (0, 1). \\ \Rightarrow f'(c_1) = 4. \end{array}$$

$$\frac{f(2) - f(1)}{2 - 1} = f'(c_2)$$

$$\Rightarrow 2 - 4 = f'(c_2) \Rightarrow f'(c_2) = -2. \\ c_2 \in (1, 2).$$

Darboux's th gives $f'(c) = 0$.

→ only one point vanishing f' .
2 points not necessarily.

3. Let $f(x)$ be a polynomial with real coefficients of degree 2025. Then

- (A) There exists $b \in \mathbb{R}$ such that $f(b) = f'(b)$.
- (B) There exists $b \in \mathbb{R}$ such that $f(b) = 0$.
- (C) There exists $b \in \mathbb{R}$ such that $f'(b) = 0$.
- (D) $f(b) = f'(b)$ for every $b \in \mathbb{R}$.

TRUE:								FALSE:		
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True B, A,
False C, D.

odd degree

$$g(x) = f(x) - f'(x)$$

→ odd degree

$f'(x)$ has degree 2024
→ even.

$$f(x) = x^{2025} + x.$$

$$f'(x) = 2025 x^{2024} + 1 > 0.$$

4. Determine which of the following statements are True and which are False.

(A) $(n+1)\log\left(1+\frac{1}{n}\right) \rightarrow 1$ as $n \rightarrow \infty$.

(B) $n^2\log\left(1+\frac{1}{n}\right) \rightarrow 1$ as $n \rightarrow \infty$.

(C) $(3n+1)\log\left(1+\frac{1}{3n}\right) \rightarrow 1$ as $n \rightarrow \infty$.

(D) $9n^2\log\left(1+\frac{1}{3n}\right) \rightarrow 1$ as $n \rightarrow \infty$.

TRUE:							FALSE:		
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True A, C
False B, D.

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = \lim_{x \rightarrow 0} \frac{1}{1+x} = 1.$$

So, near zero
 $\log(1+x) \sim x$.

$$(n+1) \log(1 + \frac{1}{n}) \\ \sim (n+1) \frac{1}{n} \rightarrow 1.$$

$$n^2 \log(1 + \frac{1}{n}) \\ \sim n^2 \cdot \frac{1}{n} = n \rightarrow \infty.$$

5. Determine which of the following statements are True and which are False.
- (A) For $0 < a < b < \infty$, $(a^{1/n} + b^{1/n})^n \rightarrow b$.
 - (B) If $u_n \rightarrow u > 0$ and $v_n \rightarrow v$, then $(u_n)^{v_n} \rightarrow u^v$.
 - (C) $\frac{5^{3n}}{6^{2n}} \rightarrow 0$ as $n \rightarrow \infty$.
 - (D) For any sequence $\{u_n\}_{n \geq 1}$, the sequence $x_n := \sup\{u_n, u_{n+1}, u_{n+2}, \dots\}$ is monotonically decreasing, whenever x_n is finite.

TRUE:								FALSE:				
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True B, D

False A, C

$$\left. \begin{array}{l} a^{1/n} \rightarrow 1 \\ b^{1/n} \rightarrow 1 \end{array} \right\} \Rightarrow (a^{1/n} + b^{1/n})^n \rightarrow 2^n \rightarrow \infty.$$

$$(u_n)^{v_n} = e^{v_n \log u_n}.$$

$$v_n \log u_n \rightarrow v \log w$$

$$\text{So, } (u_n)^{v_n} \rightarrow u^v.$$

$$\{u_n, u_{n+1}, \dots\} \supseteq \{u_{n+1}, \dots\}$$

$$\text{So, } x_{n+1} \leq x_n.$$

✓

$$\frac{5^{3n}}{6^{2n}} = \left(\frac{5^3}{6^2} \right)^n = \left(\frac{125}{36} \right)^n \rightarrow \infty$$

$$\text{as } \frac{125}{36} > 1.$$

6. A sequence $\{u_n\}_{n \geq 1}$ is defined by $u_1 > 0$ and

$$u_{n+1} = \sqrt{10 + 3u_n}, \quad n \geq 1.$$

Then

- (A) $\{u_n\}_{n \geq 1}$ is monotonically increasing if $0 < u_1 < 5$.
- (B) $\{u_n\}_{n \geq 1}$ is monotonically decreasing if $u_1 > 5$.
- (C) $\{u_n\}_{n \geq 1}$ is convergent for every $u_1 > 0$.
- (D) $\lim_{n \rightarrow \infty} u_n = 5$ for every $u_1 > 0$.

$$\begin{aligned}
 u_{n+1} - u_n &= \sqrt{10+3u_n} - \sqrt{10+3u_{n-1}} \\
 &= \frac{\cancel{10} + 3u_n - \cancel{10} - 3u_{n-1}}{\sqrt{10+3u_n} + \sqrt{10+3u_{n-1}}} \\
 &= \frac{3(u_n - u_{n-1})}{\sqrt{10+3u_n} + \sqrt{10+3u_{n-1}}}
 \end{aligned}$$

$$\begin{aligned}
 u_{n+1} > u_n &\Rightarrow \sqrt{10+3u_n} > u_n \\
 &\Rightarrow 10 + 3u_n - u_n^2 > 0 \\
 &\Rightarrow (5 - u_n)(u_n + 2) > 0
 \end{aligned}$$

So, If $5 > u_n \dots 5 > u_1$ then $\uparrow$
 $5 < u \downarrow$

$$u_n \rightarrow L \Rightarrow L^2 = 10 + 3L \Rightarrow L = 5 \checkmark / -2.$$

$$u_2 = \sqrt{10+3u_1} \Rightarrow u_2^2 = 10 + 3u_1$$

$$\begin{aligned}
 \text{If } u_1 < 5 &\Rightarrow u_2^2 < 25 \\
 &\Rightarrow u_2 < 5
 \end{aligned}$$

$\dots u_n < 5$ so, $\uparrow$ and bdd
 so, Converged

$$u_1 = 5 \Rightarrow u_2 = 5, u_3 = 5 \dots \rightarrow 5.$$

A B C D True.



Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$|f(x) - f(y)| \leq C|x - y|^2$$

for some constant $C > 0$ and for all $x, y \in \mathbb{R}$.

Which of the following must be true?

- (A) f is continuous but need not be differentiable.
 (B) f is differentiable and $f'(x)$ is bounded.
 (C) f is a constant function.
 (D) f is necessarily a linear function.

TRUE:				FALSE:		
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$$0 \leq \lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{x \rightarrow y} c |x - y| = 0$$

$$\Rightarrow \lim_{x \rightarrow y} \frac{f(x) - f(y)}{x - y} = 0.$$

So, f is diff on all $x \in \mathbb{R}$.

$$f' = 0 \quad \forall x \in \mathbb{R}.$$

$\Rightarrow f$ is constant.

Say $f(x) = 2$.

So, not required linear.

True B, C,

False A, D.

Suppose f is defined and differentiable for every $x > 0$, and

$$f'(x) \rightarrow 0 \quad \text{as } x \rightarrow +\infty.$$

Define

$$g(x) = f(x+1) - f(x).$$

Which of the following statements is true?

- (A) $g(x) \rightarrow 0$ as $x \rightarrow +\infty$.
- (B) $g(x) \rightarrow 1$ as $x \rightarrow +\infty$.
- (C) $g(x)$ need not converge to 0.
- (D) $g(x) \rightarrow f'(x)$ as $x \rightarrow +\infty$.

TRUE:							FALSE:		
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$$\begin{aligned}
 g(x) &= f(x+1) - f(x) \\
 \text{(MVT)} \quad &= f'(c_x) \quad \text{for some } c_x \in (x, x+1). \\
 &\downarrow \\
 &0 \quad \text{for } x \rightarrow +\infty.
 \end{aligned}$$

So, A true.

B false $\rightarrow$ conv to 0 not 1.

C false as $g(x)$ conv. to 0.

$$g(x) = \underbrace{f'(c_x) - f'(x)} + f'(x)$$

we do not know

$$\begin{aligned}
 &f'(c_x) - f'(x) \\
 &\rightarrow 0 \quad \text{for } x \rightarrow +\infty \text{ or not.}
 \end{aligned}$$

True A

False B C D.



Let $f : (-1, 1) \rightarrow \mathbb{R}$ be continuous and suppose that

$$f(x) = f(x^2)$$

for all $x \in (-1, 1)$.

Which of the following must be true?

- (A) f is even.
- (B) f is monotone on $(-1, 1)$.
- (C) f is constant on $(-1, 1)$.
- (D) $f(0) = 0$.

TRUE:							FALSE:		
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$$f(x) = f(x^2).$$

$$f(-x) = f((-x)^2) = f(x^2) = f(x)$$

So, f is even function.

$$\text{Let } x_0 \in (-1, 1).$$

Then

$$f(x_0) = f(x_0^2) = f(x_0^4)$$

=

$$\vdots = f(x_0^{2^n})$$

$$x_0^{2^n} \rightarrow 0 \text{ as } |x_0| < 1.$$

f is given continuous so,

$$f(x_0^{2^n}) \rightarrow f(0).$$

$$\text{So, } \forall x \in (-1, 1)$$

$$f(x) \rightarrow f(0).$$

$\Rightarrow f$ is constant function.

$\Rightarrow$ Yes monotone, (as strictly not mentioned)

$\Rightarrow f(x) = 2$ will work.

True A B C

False D

~~_____ X _____~~